A Project Stage-1
Report On

MEDIGRAPH AI: A GRAPHRAG FRAMEWORK FOR HALLUCINATION-RESISTANT MEDICAL DECISION SUPPORT

Submitted to

Kommuri Pratap Reddy Institute of Technology (Autonomous)

(Affiliated to JNTUH, Approved by AICTE & Accredited by NAAC & NBA)

In partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

Computer Science and Engineering

By

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CERTIFICATE

This is to certify that the Project Stage-1 entitled **"MEDIGRAPH AI:A GRAPHRAG FRAMEWORK FOR HALLUCINATION- RESISTANT MEDICAL DECISION SUPPORT "** being submitted by **S. Sravani (23RA1A05E6), CH. Sri Vasavi (23RA1A05F4), D. Sruthi (23RA1A05F5), S. Gnaneshwar (24RA1A0505)** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to the Kommuri Pratap Reddy Institute of Technology (Autonomous)(Affiliated to JNTUH, Approved by AICTE & Accredited by NAAC & NBA) is a record of confined work carried out by them under my guidance and supervision..

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DECLARATION

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This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this thesis have not been submitted to any other University and Institute for the award of any Degree or Diploma.

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IM3	To contribute to advancement of Engineering & Technology that would help to satisfy the societal needs.
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ABSTRACT

Artificial Intelligence and Large Language Models are rapidly transforming healthcare by supporting medical information retrieval, question answering, and clinical decision-support applications. However, conventional Large Language Models may generate incorrect, incomplete, or hallucinated information because their responses are not always directly grounded in reliable medical sources. Such limitations can reduce the trustworthiness of AI systems in healthcare, where accurate and evidence-based information is essential. To address this challenge, MediGraph AI is proposed as a Graph Retrieval-Augmented Generation framework for hallucination-resistant medical decision support. The proposed system combines Knowledge Graphs, Retrieval-Augmented Generation, Natural Language Processing, and Large Language Models to provide reliable and context-aware medical responses.When a user provides a medical query, the system processes the query, identifies relevant medical entities, and retrieves the most appropriate information and relationships from the Knowledge Graph. The framework can support medical information retrieval, disease-related knowledge exploration, symptom analysis, treatment information, and evidence-based decision support. MediGraph AI demonstrates the potential of GraphRAG technology to reduce hallucinations, improve contextual understanding, enhance explainability, and increase the reliability of Large Language Model-based medical decision-support systems.

Key Words: *MediGraph AI, GraphRAG, Knowledge Graph, Large Language Models, Retrieval-Augmented Generation, Medical Artificial Intelligence, Medical Decision Support, Hallucination Reduction, Explainable AI, Natural Language Processing, Healthcare AI, Medical Knowledge Graph, Evidence-Based Retrieval, Clinical Decision Support, Reliable AI.*

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1. INTRODUCTION

The rapid advancement of Artificial Intelligence (AI), particularly Large Language Models (LLMs), has created new opportunities in the healthcare industry. AI-powered systems can assist healthcare professionals in analyzing medical information, understanding clinical conditions, retrieving relevant medical knowledge, and supporting decision-making. However, conventional LLM-based systems face significant challenges in medical applications, including hallucinations, inaccurate information generation, limited contextual understanding, and a lack of reliable evidence to support their responses. Since medical decisions directly impact patient health and safety, providing accurate, consistent, and evidence-based information is essential.

MediGraph AI is proposed as an intelligent medical decision-support framework that integrates Retrieval-Augmented Generation **(**RAG**)** with Graph Retrieval-Augmented Generation (GraphRAG**)** to improve the reliability and contextual accuracy of AI-generated medical responses. The framework combines the capabilities of Large Language Models, knowledge graphs, and semantic information retrieval to provide structured, context-aware, and evidence-supported medical information.Unlike traditional RAG systems that primarily retrieve relevant information from document-based knowledge sources, MediGraph AI utilizes a knowledge graph to represent relationships among medical concepts such as diseases, symptoms, medications, treatments, diagnostic procedures, and clinical guidelines. By identifying connections between these entities, the system can retrieve more meaningful and contextually relevant information. This graph-based retrieval mechanism helps improve the understanding of complex medical queries and reduces the possibility of generating unsupported or misleading responses.The proposed framework follows a structured workflow involving medical knowledge extraction, knowledge graph construction, semantic retrieval, graph-based contextual reasoning, and response generation. When a user submits a medical query, the system identifies relevant medical entities, retrieves associated information from the knowledge graph and supporting knowledge sources, and provides the retrieved context to the language model. The LLM then generates a response grounded in the available medical evidence, improving transparency and reducing the risk of hallucinations.

1.1.1 Background

Healthcare is among the most information-intensive domains in the world. Medical knowledge doubles in a matter of years, clinical guidelines are revised regularly, and a single clinician must reconcile patient history, laboratory findings, imaging, drug interactions and evidence from the literature within minutes. Digital tools that organise this information and present it at the point of care therefore have the potential to improve diagnostic accuracy, reduce adverse drug events and shorten the time needed to reach a safe decision.

The recent arrival of large language models (LLMs) has changed expectations of what

such tools can do. Models trained on web-scale text can pass licensing-style medical examinations, explain clinical concepts in plain language and draft summaries of patient records. Their usefulness, however, is limited by a fundamental property of how they work: an LLM generates the most probable continuation of a text, not a verified statement of fact. When the model lacks knowledge, or when the relevant knowledge is rare or recent, it can still produce a confident and well-formed answer that is simply wrong. This behaviour is known as hallucination.

In everyday applications a hallucination is an inconvenience. In medicine it is a patient-safety risk. A fabricated drug dose, an invented clinical trial or an overlooked contraindication can cause real harm, and clinicians cannot be expected to verify every sentence of an AI-generated answer manually. For AI to be adopted responsibly in clinical settings, answers must be grounded in trusted sources, traceable to evidence and honest about uncertainty.

1.1.2 Motivation

Several observations motivate the work. Medical knowledge is naturally relational: diseases cause symptoms, drugs treat diseases, drugs interact with other drugs, and tests confirm diagnoses. A graph is the most faithful data structure for such knowledge, and graph traversal gives a transparent reasoning path that a clinician can inspect. Recent GraphRAG research has shown that organising a corpus as a graph with hierarchical community summaries improves answers to questions that require synthesis across many documents. Finally, in India and other regions with a high patient-to-doctor ratio, reliable decision-support tools can extend the reach of scarce expertise, provided that they are safe, explainable and compliant with data-protection law.

1.2 Problem Statement

Existing LLM-based medical assistants suffer from three connected weaknesses. First, they rely on parametric memory, so their knowledge is frozen at training time and cannot be audited. Second, conventional retrieval-augmented generation retrieves flat text chunks by semantic similarity; this works for single-fact lookup but fails on questions that require following chains of relationships, such as identifying which first-line therapy for a condition is unsafe for a patient who also has renal impairment. Third, almost no deployed system verifies the generated answer after it is produced, so unsupported statements reach the user unchallenged.

The problem addressed by this project can therefore be stated as follows: to design a medical decision-support framework that answers clinical queries using structured, traceable knowledge, reasons across multiple relationships, and detects and suppresses unsupported statements before they reach the clinician.

The main objective of MediGraph AI is to build a hallucination-resistant medical question-answering and decision-support framework. The specific objectives are listed below.

- To construct a medical knowledge graph from trusted ontologies, clinical guidelines and drug references, with provenance recorded for every fact.
- To implement hybrid retrieval that combines graph traversal, community-level summaries and dense vector search for multi-hop clinical questions.
- To design a grounded generation strategy in which the LLM may answer only from retrieved evidence.
- To develop a claim-level verification module that checks each statement of a draft answer against the graph and flags unsupported content.
- To return every answer with citations, a reasoning path and a calibrated confidence score, and to abstain when evidence is insufficient.
- To define an evaluation protocol that measures faithfulness, correctness, citation quality and safety, and compares MediGraph AI against baseline LLM and vector-RAG systems.

The framework is scoped as a clinician-facing decision-support aid for question answering over diagnosis support, treatment options, drug information, drug interactions and contraindications. It operates on curated, licensed or openly available knowledge sources. It does not make autonomous clinical decisions, does not prescribe, and is not intended for direct use by patients as a substitute for professional care. Image-based diagnosis, real-time monitoring and electronic health record (EHR) integration are treated as future extensions, although the architecture is designed so that they can be added.

2. LITERATURE SURVEY

2.1 Large Language Models in Healthcare

Transformer-based language models (Vaswani et al., 2017) have become the foundation of modern natural language processing. Domain studies have shown that large models encode a substantial amount of clinical knowledge: Singhal et al. (2023) evaluated Flan-PaLM and the instruction-tuned Med-PaLM on multiple-choice benchmarks such as MedQA and PubMedQA and on consumer health questions, and reported that performance improved with scale and with instruction prompt tuning, while also documenting gaps against clinician answers in areas such as potential harm and evidence use. Other work has trained specialised biomedical models and evaluated general-purpose models on licensing examinations. These results demonstrate potential but also expose limits. Benchmark accuracy does not guarantee safe behaviour in open-ended clinical use, the knowledge held in model weights cannot be traced to a source, and updating it requires expensive retraining. These limitations motivate architectures that keep knowledge outside the model, in a store that can be inspected, versioned and corrected.

2.2 Hallucination in Language Models

Ji et al. (2023) surveyed hallucination in natural language generation and distinguished intrinsic hallucination, where the output contradicts the source input, from extrinsic hallucination, where the output contains content that cannot be verified from the source. Huang et al. (2023) extended the discussion to LLMs and described factuality hallucination and faithfulness hallucination as two broad categories. In the medical domain the following forms are especially important. \

Table 2.1: Types of hallucination relevant to medicine

Type	Description	Clinical example
Factual fabrication	Invents entities, studies or numbers that do not exist	Citing a non-existent trial to support a drug choice
Outdated knowledge	States information superseded by newer guidance	Recommending a withdrawn first-line therapy
Contradiction	Output conflicts with the supplied context or itself	Stating a drug is safe, then listing it as contraindicated
Omission	Leaves out a critical safety fact	Not mentioning a major drug–drug interaction
Over-generalisation	Applies a fact to a population where it does not hold	Adult dosing advice given for a paediatric query
Unsupported	Draws a conclusion with no	Linking a symptom cluster to

inference	evidential link	a rare disease without basis

Mitigation strategies fall into three families: training-time methods such as fine-tuning and alignment, inference-time methods such as prompting, decoding constraints and self-consistency checks, and knowledge-grounding methods that supply external evidence. MediGraph AI belongs to the third family but adds a post-generation verification layer that belongs to the second.

2.3 Retrieval-Augmented Generation

Lewis et al. (2020) introduced retrieval-augmented generation, in which a generator conditions on passages retrieved from an external corpus by a dense retriever. RAG improves factual grounding, allows knowledge to be updated by changing the corpus, and offers a natural place to attach citations. Gao et al. (2023) surveyed the field and described the progression from naive RAG to advanced and modular RAG, with pre-retrieval query rewriting, post-retrieval re-ranking and iterative retrieval.

In medicine, Xiong et al. (2024) benchmarked RAG systems on medical question answering (the MIRAGE benchmark) and found that retrieval can substantially improve the accuracy of LLMs, while also showing that results depend on the choice of corpus, retriever and number of retrieved snippets. Self-RAG (Asai et al., 2023) trains a model to decide when to retrieve and to critique its own generations with reflection tokens, which illustrates the value of explicit self-assessment.

Limitations of Flat Vector RAG

Chunks are retrieved independently, so relationships that span documents are lost.

Similarity does not imply relevance to the clinical question; negation and dosage details are easily missed.

There is no global view of the corpus, so questions that require summarising many sources perform poorly.

Retrieved passages are not verified against the generated answer, so ungrounded sentences can still appear.

2.4 Knowledge Graphs in Medicine

A knowledge graph represents facts as typed nodes connected by labelled relationships (Hogan et al., 2021). Medicine has a rich tradition of structured knowledge. The Unified Medical Language System (UMLS) integrates many biomedical vocabularies and provides concept identifiers that link terms across sources (Bodenreider, 2004). SNOMED CT offers a comprehensive clinical terminology, ICD supports disease classification, RxNorm normalises drug names, and DrugBank and similar resources describe drug targets and interactions. Using

these standards as the backbone of the graph gives MediGraph AI consistent identifiers, synonym handling and interoperability.

GraphRAG extends RAG by building a graph from the corpus and retrieving over that structure. Edge et al. (2024) proposed a pipeline in which an LLM extracts entities and relationships from text chunks, the resulting graph is partitioned into communities using the Leiden algorithm (Traag et al., 2019), and each community is summarised; query-time answers can then be assembled from community summaries to address global questions that flat RAG cannot answer. Pan et al. (2024) described a roadmap for unifying LLMs and knowledge graphs, in which graphs enhance LLMs with explicit knowledge and LLMs help build and complete graphs.

Graphs support explicit multi-hop reasoning, are auditable because each edge can carry a source, and can be updated incrementally. Their main cost is construction: extracting reliable entities and relations from text at scale requires careful pipelines, ontology alignment and quality control.

2.5 Graph-Based Retrieval-Augmented Generation

The Louvain method (Blondel et al., 2008) and Leiden algorithm are widely used for community detection and are suited to the hierarchical summarisation that GraphRAG needs. MediGraph AI adopts this idea but changes two things for the medical setting: the graph is seeded with standard ontologies rather than being extracted purely by an LLM, and the graph is used again after generation to verify the answer.

2.6 Hallucination Detection and Factuality Evaluation

Evaluating whether a long answer is faithful requires more than comparing it with a reference. FActScore (Min et al., 2023) decomposes a generated text into atomic facts and measures the proportion supported by a knowledge source. RAGAS (Es et al., 2023) proposes reference-free metrics for RAG pipelines, including faithfulness, answer relevance and context relevance. These ideas directly inspire the claim extraction and scoring modules of MediGraph AI, which apply atomic-claim checking not only for evaluation but also at inference time as a safety filter.

2.7 Research Gap

The literature shows that LLMs are knowledgeable but unreliable, that RAG improves grounding, and that graph structure improves multi-hop and global reasoning. What remains underexplored is a medically specialised GraphRAG design that (i) builds its graph on standard clinical ontologies with full provenance, (ii) verifies each claim of the generated answer against that graph before release, and (iii) treats abstention as a first-class, measurable behaviour. MediGraph AI is proposed to fill this gap.

3. EXISTING SYSTEM

3.1 Overview of Existing Systems

Clinicians today consult drug compendia, guideline portals, literature databases and, increasingly, general-purpose chatbots. Each source has a different interface and terminology, and no single tool reconciles them. General chatbots are convenient but offer no guarantee of correctness. Clinical decision support systems embedded in hospital software typically rely on rule-based alerts, which are accurate for narrow cases such as drug–drug interaction warnings but are not able to answer open-ended questions and often cause alert fatigue.

Drawbacks of the Existing System

Information is scattered across many sources and formats.

LLM chatbots hallucinate and cannot cite verifiable evidence.

Vector-based RAG cannot reliably chain relationships across documents.

Rule-based alerts are rigid and do not explain their reasoning in natural language.

There is no standard way to measure or enforce the faithfulness of AI answers

3.2 Comparative Analysis of Existing Approaches

Table 3.1: Comparison of approaches to medical question answering

Approach	Strengths	Weaknesses	Hallucination control
Plain LLM	Fluent, broad, no infrastructure	No sources, stale knowledge	None
Fine-tuned medical LLM	Better domain accuracy	Costly to update, not traceable	Weak
Vector RAG	Grounded in documents, updatable	Weak multi-hop reasoning, no verification	Moderate
Knowledge-graph QA	Precise, explainable paths	Rigid, limited language fluency	Strong but narrow
GraphRAG	Multi-hop and global synthesis	Construction cost, no medical safety layer by default	Moderate to strong

MediGraph AI	Graph plus vector retrieval, provenance, verification, abstention	Needs curated sources and expert validation	**Strong (pre- and post-generation)**

3.3 Illustrative Failure Scenarios

The weaknesses described above become concrete when they are applied to realistic clinical questions. Table 3.2 lists representative scenarios in which existing approaches fail and indicates the underlying reason.

Table 3.2: Failure scenarios of existing approaches

Clinical question	Typical failure	Root cause
First-line drug for hypertension in a diabetic patient with kidney disease	Plain LLM names a drug class without noting renal cautions	No access to patient-specific contraindications
Is drug A safe together with drug B?	Vector RAG returns separate passages on A and B and misses the interaction passage	Relationship between A and B is not modelled
Recent guideline change for a disease	LLM answers with superseded recommendation	Static training data, no versioned knowledge
Summarise treatment options across all available guidelines	RAG returns only a few similar chunks, giving a partial picture	No global view of the corpus
Dose adjustment for a child	Adult dose stated confidently	Population qualifiers lost in chunking or in model memory

In each scenario the answer sounds plausible, which is precisely what makes the failure dangerous. A clinician who is short of time may accept the statement without checking it.

The system must therefore not only produce better answers but also be able to show why an answer is trustworthy and decline to answer when it cannot.

3.4 Need for a New Approach

The analysis leads to four needs. Knowledge must be stored in a structured, versioned form so that it can be audited and corrected. Retrieval must follow relationships rather than only textual similarity so that multi-hop questions are answered completely. Generation must be restricted to the retrieved evidence. Finally, an independent verification step must examine the generated answer, because instructing a language model to behave is not a guarantee that it will. MediGraph AI is designed

4. PROPOSED SYSTEM

4.1 Overview of the Proposed System

MediGraph AI proposes a pipeline in which a knowledge graph is the single source of truth. A clinician asks a question in natural language. The system identifies medical entities, retrieves the connected sub-graph and relevant community summaries, and passes this structured evidence, together with supporting text passages, to an LLM under a strict grounding instruction. The draft answer is decomposed into atomic claims and each claim is verified against the graph. Verified answers are returned with citations and reasoning paths; failed claims are removed, regenerated or replaced with an explicit statement that the evidence is insufficient.

Table 4.1: Existing versus proposed system

Aspect	Existing systems	MediGraph AI
Knowledge source	Model memory or flat documents	Typed knowledge graph with provenance
Reasoning	Single-hop similarity	Multi-hop graph traversal and community summaries
Verification	None or manual	Automatic claim-level verification
Explainability	Opaque	Citations and visible reasoning path
Uncertainty	Rarely expressed	Confidence score and abstention
Update method	Retraining or re-indexing	Incremental graph update with version history

4.2 Functional Requirements

Table 4.3: Non-functional requirements

Category	Requirement
Accuracy and safety	Target faithfulness above 95 percent on the evaluation set; zero tolerance for unflagged unsupported drug-dose statements
Performance	Median end-to-end response time under 8 seconds; retrieval stage under 1.5 seconds
Scalability	Support graphs of several million nodes and horizontal scaling of the API tier
Explainability	Every claim linked to at least one source node or passage
Security	Encryption in transit and at rest, role-based access

	control, audit trail
Privacy	No identifiable patient data stored by default; minimal data retention
Maintainability	Modular components with versioned schemas and automated tests
Usability	Clear interface designed for clinicians, accessible on desktop and tablet
Availability	Target availability of 99 percent for the clinician-facing service

4.4 Hardware and Software Requirements

Table 4.4: Development and deployment environment

Component	Specification
Processor	Minimum 8-core CPU (16-core recommended for ingestion)
Memory	Minimum 32 GB RAM (64 GB for graphs above one million nodes)
GPU	Optional 16 GB GPU for local embedding and open-source LLM inference
Storage	Minimum 500 GB SSD
Operating system	Ubuntu 22.04 LTS or later (Windows 11 for development)
Languages and runtime	Python 3.11, Node.js 20
Graph database	Neo4j 5.x or an equivalent property-graph store
Vector index	FAISS or a vector extension of the graph database
Frameworks	FastAPI, LangChain or LlamaIndex, spaCy / scispaCy, PyTorch, React
Containers	Docker and Docker Compose; Kubernetes for production

4.5 Feasibility Study

Technical Feasibility

All required technologies are mature and available as open source or through commercial APIs. Graph databases handle multi-hop traversal efficiently, vector search is a commodity capability, and LLMs can be accessed through hosted APIs or run locally in quantised form. The principal technical risk is graph construction quality, which is mitigated by ontology seeding and human review of high-risk relations.

Economic Feasibility

Open-source tooling keeps infrastructure costs modest. The main recurring costs are LLM inference and storage, which can be controlled through caching, a smaller verification model and retrieval of only the necessary sub-graph. Licensing of terminologies such as SNOMED CT is available at no cost for many research uses and should be confirmed for the deployment country.

Operational Feasibility

The system fits the existing clinical workflow of asking a question and reading a concise answer. Because answers include sources and reasoning paths, clinicians can verify them rapidly, which supports adoption. Knowledge administrators need a simple review console, which is included in the design.

Legal and Ethical Feasibility

Using only de-identified or synthetic patient data in development, applying the India Digital Personal Data Protection Act 2023 and, where relevant, HIPAA principles, and presenting the tool as decision support rather than diagnosis keeps the project within acceptable regulatory bounds. Section 4.9 expands on these aspects.

4.6 Actors and Use Cases

Table 4.5: Actors and their principal interactions

Actor	Role	Main use cases
Clinician	Primary user	Ask clinical question, view evidence, check interactions, give feedback
Medical student	Secondary user	Learn disease–drug relationships with cited explanations
Knowledge administrator	Maintains content	Ingest sources, review extracted relations, approve versions
Clinical reviewer	Domain expert	Validate high-risk facts, evaluate answers

System administrator	Operations	Manage users, monitor performance, review audit logs

4.7 Risk Analysis

Table 4.6: Key risks and mitigation

Risk	Impact	Mitigation
Incorrect relation extracted into graph	High	Ontology seeding, confidence thresholds, expert review of safety-critical edges
Outdated guideline in graph	High	Versioning, validity dates on facts, scheduled refresh
LLM ignores grounding instruction	High	Post-generation claim verification and regeneration
Entity linking error	Medium	Hybrid lexical and embedding linking, ambiguity prompts to user
Over-reliance by clinicians	High	Prominent disclaimers, confidence display, abstention behaviour
Privacy breach	High	De-identification, encryption, access control, audit logs
Latency too high	Medium	Caching, bounded hop depth, asynchronous verification of low-risk claims

4.8 Evaluation Plan

A hallucination-resistant system must be evaluated on whether its answers are faithful and safe, not only on whether they are correct. The evaluation plan therefore measures several complementary properties and compares MediGraph AI with simpler baselines under identical conditions.

Datasets

- Public medical question-answering sets such as MedQA, MedMCQA and PubMedQA for answer correctness.
- A curated set of multi-hop clinical questions about drug interactions, contraindications and guideline-based treatment, written and validated by clinical experts.
- An adversarial set containing questions with false premises, non-existent drugs and out-of-scope requests, used to test abstention and resistance to fabrication.
- Synthetic patient scenarios with stated age, conditions and medications to test patient-context handling.

Metrics

Table 4.7: Evaluation metrics

Metric	Definition	Goal
Faithfulness	Share of atomic claims in the answer that are supported by the supplied evidence (after FActScore and RAGAS)	Maximise
Answer correctness	Agreement of the final answer with the reference answer or expert judgement	Maximise
Citation precision and recall	Share of citations that truly support the claim, and share of claims that carry a valid citation	Maximise
Hallucination rate	Share of answers containing at least one unsupported or contradicted claim after verification	Minimise
Safety-critical error rate	Answers with an incorrect dose, missed contraindication or missed interaction	Minimise (target zero)
Abstention quality	Share of unanswerable questions correctly declined, and share of answerable questions wrongly declined	High / low respectively
Calibration error	Gap between stated confidence and observed accuracy	Minimise
Latency	Median and 95th-percentile end-to-end response time	Within targets of Table 4.3

Baselines and Ablations

MediGraph AI is compared with four baselines: the same LLM without retrieval, the LLM with vector RAG, the LLM with graph retrieval but no verification, and the full system. Ablation studies remove one component at a time, namely community summaries, patient-context boosting, the structured verification check, the textual entailment check and the abstention policy, to quantify the contribution of each. All systems use the same LLM, corpus and prompts wherever possible so that differences can be attributed to the architecture

Expert Review

Automated metrics are complemented by blinded review of a sample of answers by qualified clinicians, who rate correctness, completeness, safety and usefulness on a five-point scale. Disagreements are resolved by discussion, and inter-rater agreement is reported.

Reporting Template

Table 4.8: Template for reporting experimental results (to be completed after experiments)

System	Faithfulness	Correctness	Hallucination rate	Safety errors
LLM only	to be measured	to be measured	to be measured	to be measured
LLM + vector RAG	to be measured	to be measured	to be measured	to be measured
LLM + graph retrieval	to be measured	to be measured	to be measured	to be measured
MediGraph AI (full)	to be measured	to be measured	to be measured	to be measured

4.9 Security, Privacy, Ethics and Compliance

Medical software carries obligations beyond functional correctness. The design incorporates the following measures from the start rather than adding them later.

Table 4.9: Security, privacy and ethical safeguards

Area	Safeguard
Data protection	Compliance with the Digital Personal Data Protection Act, 2023 of India; HIPAA principles considered where applicable; data minimisation and purpose limitation
De-identification	Development and evaluation use synthetic or de-identified data; patient context is processed in memory and not stored unless explicitly enabled with consent
Access control	Role-based access for clinicians, reviewers and administrators; multi-factor authentication for administrators
Encryption	TLS for all traffic; encryption at rest for databases and object storage; managed secrets
Prompt-injection defence	Ingested documents are treated as data; retrieved text is delimited and never executed as instructions; outputs are verified
Auditability	Immutable logs of queries, evidence, claims and decisions, enabling retrospective review of any answer
Transparency	Citations, confidence and abstention are always displayed; limits of the system are stated in the interface
Fairness	Evaluation stratified by age, sex and condition prevalence; review of sources for population bias
Human oversight	The system supports and never replaces clinical judgement; no autonomous orders are issued
Regulatory awareness	Intended-use statement kept consistent with decision-support classification; any future clinical deployment would require regulatory assessment and clinical validation

The system is not a medical device in its current research form and must not be used to make treatment decisions without review by a qualified clinician.

4.10 Technology Stack and Implementation Outline

Table 4.10: Proposed technology stack

Layer	Technology	Reason for selection
Frontend	React, TypeScript, a graph visualisation library	Responsive interface and interactive reasoning path
API and orchestration	Python 3.11, FastAPI, LangChain or LlamaIndex	Rich ecosystem and asynchronous support
Graph store	Neo4j with Cypher queries	Mature property-graph database with graph algorithms
Vector and lexical search	FAISS or pgvector; BM25 via OpenSearch or a Python library	Hybrid retrieval with exact-term matching

The following Cypher query illustrates graph retrieval for a treatment question. It finds drugs that treat a condition and excludes those with a contraindication that matches a stated patient attribute.

```
MATCH (d:Disease {cui: $disease})<-[t:TREATS]-(drug:Drug)

OPTIONAL MATCH (drug)-[:HAS_CONTRA]->(c:Contraindication)

WHERE c.condition IN $patient_conditions

WITH drug, t, collect(c) AS blocking

WHERE size(blocking) = 0

MATCH (t)-[:SUPPORTED_BY]->(s:Source)

RETURN drug.name, t.evidence_grade, s.title, s.version

ORDER BY t.evidence_grade DESC LIMIT 10
```

The grounded prompt given to the generator has the following structure, with the evidence block inserted at run time.

4.11 Expected Outcomes, Limitations and Future Scope\

Expected Outcomes

- Answers that are traceable to named sources and displayed with citations and reasoning paths.
- A measurable reduction in unsupported claims compared with LLM-only and vector-RAG baselines.
- Better handling of multi-hop questions about interactions, contraindications and guideline-based therapy.
- Safe behaviour under uncertainty through calibrated confidence and abstention.

Limitations

- Answer quality is bounded by the coverage and accuracy of the knowledge sources and the extraction pipeline.
- Graph construction and clinical review are resource-intensive, especially for safety-critical relations.
- Verification itself relies on language models and can make errors; it reduces but cannot eliminate risk.
- Rare diseases, novel therapies and conflicting evidence are harder to represent.
- The system has not been clinically validated and is a research prototype.

Future Scope

- Integration with electronic health records through standards such as HL7 FHIR for richer patient context.
- Multimodal inputs such as laboratory reports and medical images.
- Temporal knowledge graphs that represent when a recommendation was valid and track guideline evolution.
- Multilingual support, including Indian languages such as Telugu and Hindi, for wider access.
- Active learning in which clinician feedback is used to correct graph edges and improve retrieval.

4.12 Conclusion

MediGraph AI proposes that safe medical use of language models depends less on making the model itself infallible and more on surrounding it with a trustworthy knowledge structure and an independent verifier. By organising medical knowledge as a provenance-rich graph, retrieving evidence through both graph traversal and semantic search, constraining generation to that evidence, and checking every claim before release, the framework addresses the main causes of hallucination in clinical question answering. Its answers carry citations, reasoning paths and confidence scores, and it declines to answer when evidence is inadequate. The design, algorithms and evaluation plan presented in this document provide a basis for implementation and for rigorous, reproducible assessment, with the aim of supporting clinicians while keeping them firmly in control of decisions.

5. SYSTEM ARCHITECTURE

5.1 Design Principles

The design of MediGraph AI follows five principles that are applied consistently across all components.

Grounding first. The LLM is a language interface to evidence, not a source of medical facts.

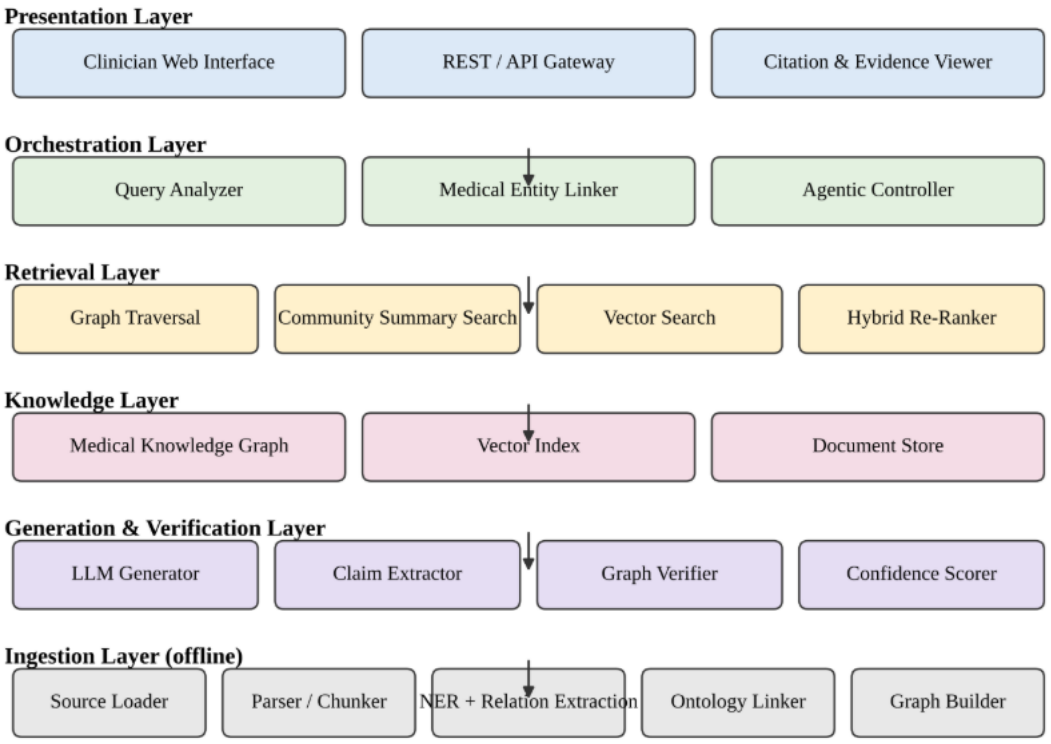
Provenance everywhere. Every node, edge and passage records where it came from, when, and with what evidence grade.

Verify after generating. Safety does not depend on the model obeying instructions; a separate component checks the result.

Fail safe. When evidence is missing, conflicting or low-confidence, the system abstains and says so.

Modularity. Each stage can be replaced or improved independently, for example a different LLM or graph store.

5.2 High-Level Architecture



MediGraph AI is organised into six layers, shown in Figure 5.1. The ingestion layer runs offline and builds the knowledge layer. The presentation, orchestration, retrieval and generation-and-verification layers run online for each query.

5.3 Description of Layers Description of Layers

5.3.1 Presentation Layer

The presentation layer offers a web interface and a REST API. The interface accepts a question and optional patient context, displays the answer with inline citation markers, shows the reasoning path as a small graph visualisation, and displays a confidence indicator. A separate administrator view supports ingestion review and monitoring

5.3.2 Orchestration Layer

The orchestration layer controls the flow of a query. The query analyser classifies the intent (for example diagnosis support, treatment, drug information, interaction check or general explanation) and rewrites the query when necessary. The medical entity linker recognises mentions of diseases, drugs and symptoms and maps them to standard concept identifiers. An agentic controller decides which retrieval strategies to run and whether a second retrieval round is required when the first evidence set is insufficient**.**

5.3.3 Retrieval Layer

Retrieval combines four mechanisms. Graph traversal expands the linked entities along clinically meaningful relationships up to a bounded number of hops. Community summary search retrieves pre-computed summaries of related groups of entities for broad questions. Vector search retrieves passages semantically close to the query. A hybrid re-ranker merges these candidates, scores them for relevance, authority and recency, and returns a compact evidence set that fits the LLM context window.

5.3.4 Knowledge Layer

The knowledge layer stores the medical knowledge graph, the vector index of passages and node descriptions, and the document store containing the original source text. Identifiers link all three, so a claim in the graph can always be traced to an exact passage.

5.3.5 Generation and Verification Layer

The LLM generator produces a draft answer under a grounding prompt. The claim extractor splits the draft into atomic claims. The graph verifier checks each claim against the evidence and the wider graph, labelling it supported, contradicted or unsupported. The confidence scorer aggregates retrieval strength, verification results and source authority into an overall score that determines whether the answer is released, regenerated or replaced by an abstention.

5.3.6 Ingestion Layer

The ingestion layer loads sources, parses and chunks them, extracts entities and relations, links them to ontologies, and writes them to the graph with provenance. It also computes embeddings, detects communities and generates summaries. It runs as a batch or incremental job and writes a new graph version only after validation checks pass.

5.4 Query Processing Flow

Figure 5.2 shows the sequence followed for each query. The loop from graph verification back to generation limits the number of regeneration attempts to a small configured value so that latency remains bounded.

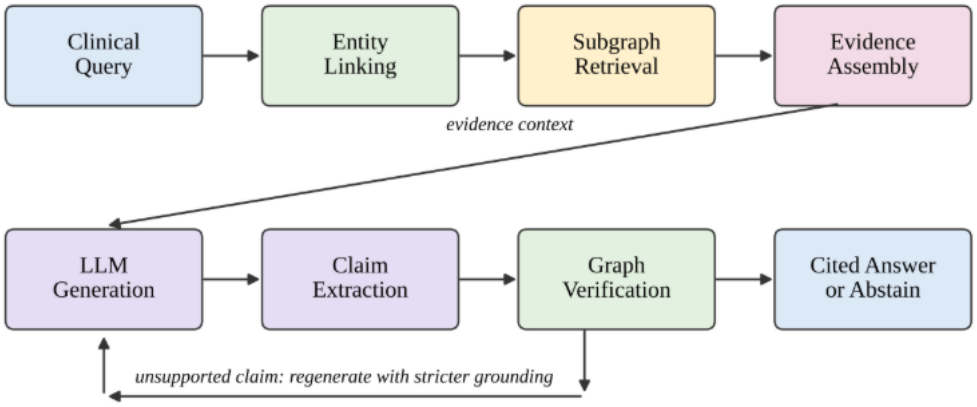


Figure 5.2: End-to-end query processing pipeline

5.5 Knowledge Graph Schema

The graph uses a property-graph model. The principal node types and relationships are shown in Figure 5.3. Disease is the central entity type because most clinical questions revolve around a condition, its manifestations, its work-up and its management.

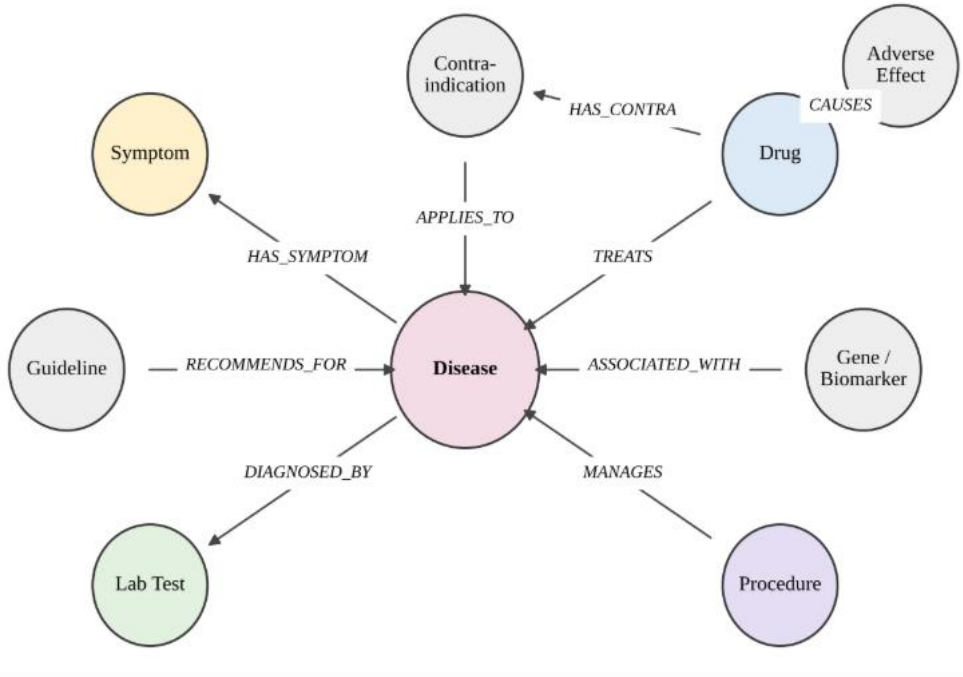


Figure 5.3: Core entity and relationship schema of the medical knowledge graph

Table 5.1: Node types and key properties

Node type	Key properties	Reference vocabulary
Disease	cui, name, synonyms, icd_code, description	UMLS, SNOMED CT, ICD
Symptom / Sign	cui, name, body_system	SNOMED CT
Drug	rxcui, name, class, route, pregnancy_category	RxNorm, DrugBank
Lab Test	loinc_code, name, reference_range, unit	LOINC
Procedure	code, name, invasiveness	SNOMED CT
Contraindication	id, condition, severity, population	Drug labels
Adverse Effect	cui, name, frequency, severity	MedDRA, drug labels
Guideline	id, title, issuer, year, evidence_grade	Society guidelines
Gene / Biomarker	symbol, name, variant	HGNC
Source	source_id, title, publisher, date, licence, version	Internal

Table 5.2: Relationship types

Relationship	From → To	Meaning
HAS_SYMPTOM	Disease → Symptom	Condition commonly presents with the symptom
DIAGNOSED_BY	Disease → Lab Test	Test supports or confirms diagnosis
TREATS	Drug → Disease	Drug is indicated for the condition
MANAGES	Procedure → Disease	Procedure is used to manage the condition
HAS_CONTRA	Drug → Contraindication	Drug should not be used under the stated condition
INTERACTS_WITH	Drug → Drug	Interaction with severity and mechanism
CAUSES	Drug → Adverse Effect	Drug may cause the effect
RECOMMENDS_FOR	Guideline → Disease	Guideline recommendation for the condition
ASSOCIATED_WITH	Gene → Disease	Genetic association

5.6 Module Design

Table 5.3: Principal software modules

Module	Responsibility	Inputs → Outputs
Ingestion Service	Load, parse, chunk and register sources	Raw documents → chunks with metadata
Extraction Service	Entities and relations from text	Chunks → candidate triples with confidence
Ontology Linker	Map mentions to concept identifiers	Mentions → canonical IDs
Graph Builder	Validate and write nodes and edges	Triples → versioned graph
Community Engine	Detect communities and summarise	Graph → hierarchy of summaries
Query Analyzer	Intent detection and rewriting	Question → structured query object
Retriever	Graph, summary and vector retrieval	Query object → ranked evidence
Generator	Grounded answer drafting	Evidence + question → draft answer
Verifier	Claim extraction and checking	Draft + graph → labelled claims
Policy Engine	Release,	Scores → final

	regenerate or abstain	response
Audit Service	Logging and traceability	All events → immutable log

5.7 Database Design

Three stores are used. The graph database holds nodes, relationships and their properties. The vector index holds embeddings for passages, node descriptions and community summaries, each with an identifier that points back to the graph or document store. A relational database holds users, roles, audit logs, feedback and evaluation records. Separating transactional records from the graph keeps the knowledge store clean and allows independent scaling.

Table 5.4: Relational tables for operations and audit

Table	Main columns
users	user_id, name, role, organisation, created_at
queries	query_id, user_id, text, intent, timestamp, graph_version
evidence_log	query_id, node_or_passage_id, rank, score
claims	claim_id, query_id, text, status, supporting_ids, score
responses	query_id, answer_text, confidence, released, abstained, latency_ms
feedback	feedback_id, query_id, rating, comment, reviewer_role

5.8 API Design

Method and path	Purpose
POST /v1/query	Submit a question and optional patient context; returns answer, claims, citations and confidence
GET /v1/evidence/{id}	Return the source passage or graph path behind a citation
POST /v1/interactions	Check a list of drugs and conditions for interactions and contraindications
POST /v1/feedback	Record clinician feedback
POST /v1/admin/ingest	Start an ingestion job for an approved source
GET /v1/admin/graph/version	List graph versions and validation reports

Table 5.5: Principal REST endpoints

5.9 User Interface Design

The interface follows a clean, low-clutter layout suited to clinical use. The main screen has a question box with an optional collapsible panel for patient context. The result area shows the answer first, followed by a verification badge (verified, partially verified or abstained), numbered citations that open the underlying passage, a compact reasoning-path graph, and an expandable list of verified claims. Warnings for high-risk items such as interactions are highlighted. A persistent notice states that the output is decision support and must be reviewed by a qualified professional.

6. MODULES

MediGraph AI is divided into functional modules that correspond to the stages of the pipeline. Table 6.1 summarises the modules and indicates the section in which each is described. The sections that follow explain the working of each module, and the last four sections give the algorithms and a worked example.

Table 6.1: Module overview

Module	Purpose	Section
M1: Knowledge Source Management	Select, license and weight trusted sources	6.1
M2: Document Preprocessing	Clean, structure and chunk documents	6.2
M3: Knowledge Extraction and Linking	Extract entities and relations, link to ontologies	6.3, 6.4
M4: Graph Construction and Quality Control	Build, validate and version the knowledge graph	6.5
M5: Community Summarisation and Indexing	Summaries, embeddings and hybrid index	6.6, 6.7
M6: Query Understanding and Retrieval	Intent, entities, patient context, hybrid retrieval	6.8, 6.9
M7: Grounded Generation	Evidence assembly and constrained answer drafting	6.10
M8: Verification and Policy	Claim checking, confidence scoring, abstention	6.11, 6.12

6.1 Module M1: Knowledge Source Management

Table 6.2: Candidate knowledge sources

Source type	Examples	Use in the system
Terminologies	UMLS Metathesaurus, SNOMED CT, ICD, LOINC, RxNorm	Backbone concepts, synonyms and identifiers
Drug references	DrugBank, regulatory drug labels, national formularies	Indications, doses, interactions, contraindications
Clinical guidelines	WHO, ICMR, specialty society guidelines	Recommendations with evidence grade
Literature	PubMed abstracts, PMC open-access articles	Supporting evidence and recent findings
Question sets	MedQA, PubMedQA, MedMCQA	Evaluation, not ingestion into the graph
Synthetic or de-identified cases	Synthetic patient records, de-identified datasets with permission	Testing patient-context handling

Each source is registered with a licence record. Sources are given an authority weight, with regulatory labels and national or international guidelines above review articles and individual studies. The weight is used by the re-ranker and the confidence scorer.

6.2 Module M2: Document Preprocessing

Documents in PDF, HTML or XML form are converted to clean text with section structure preserved. Headers, footers and navigation text are removed. Tables, which are very common in drug labels and guidelines, are converted to structured rows so that doses and conditions are not separated from their context. The text is then split into chunks of roughly 300 to 600 tokens along section boundaries with a small overlap, and each chunk is stored with its source identifier, section title, publication date and version.

6.3 Module M3: Entity and Relation Extraction

Extraction uses two complementary routes. A biomedical named-entity recognition

model, such as one from the scispaCy family, tags diseases, chemicals and anatomical terms quickly and cheaply. An LLM extraction prompt then reads each chunk with the recognised entities and returns candidate relations in a fixed JSON schema restricted to the relationship types of Table 5.2. For every candidate triple the model must also return the exact supporting sentence. Candidates whose supporting sentence cannot be found verbatim in the chunk are discarded, which removes many extraction hallucinations at the source.

6.4 Normalisation and Ontology Linking

Extracted mentions are mapped to canonical concepts. The linker first applies exact and fuzzy lexical matching against concept names and synonyms. For ambiguous mentions it computes embedding similarity between the mention context and concept descriptions and selects the best candidate above a threshold. Mentions that remain ambiguous are kept as provisional nodes and flagged for review. Drug names are normalised to ingredient level, with brand names, strengths and forms retained as properties, so that a question using a brand name reaches the same node as a guideline using the generic name.

6.5 Module M4: Graph Construction and Quality Control

Triples are merged into the graph with the following rules. Nodes with the same canonical identifier are unified. A relation seen in several independent sources receives a higher support count. Conflicting relations, such as a drug marked both indicated and contraindicated for the same population, are not silently merged but stored with a conflict flag. Every relation is attached to its source and supporting sentence through SUPPORTED_BY edges.

Quality control combines automated and human checks. Automated checks test schema validity, detect orphan nodes, verify that dose values fall in plausible ranges and compare new edges against curated reference sets. Safety-critical relation types, namely contraindications, interactions and dosing limits, are routed to a clinical reviewer when their confidence is below a threshold or when they appear in only one source. A graph version is published only after validation passes.

6.6 Module M5: Community Detection and Summarisation

To support broad questions, the graph is partitioned into communities using the Leiden algorithm applied hierarchically. A community may correspond, for instance, to the cluster of cardiovascular conditions, drugs and tests that are densely connected. For each community an LLM generates a short summary from the member nodes, their relationships and the supporting sentences, and the summary is stored with links to its members. Summaries are regenerated only for communities that changed in a new graph version, which keeps update cost low.

6.7 Embedding and Hybrid Index

Three kinds of text are embedded: source passages, node descriptions and community summaries. A biomedical or general-purpose sentence-embedding model is used, and vectors are stored in the index with identifiers. In addition, a lexical index (BM25) is maintained because exact terms such as drug names, doses and codes are often better matched lexically than semantically. At retrieval time, dense and lexical scores are fused with reciprocal rank fusion.

6.8 Module M6: Query Understanding

The query analyser performs four tasks. It classifies intent, which selects the retrieval strategy and the answer template. It extracts and links entities, using the same linker as ingestion. It extracts structured patient context, such as age group, pregnancy, renal or hepatic impairment, current medications and allergies, because these often determine the safe answer. Finally, it decides whether the question is answerable within scope; questions outside scope or requesting direct personal treatment orders are answered with guidance to consult a clinician.

6.9 Hybrid Graph Retrieval

Retrieval proceeds in four steps. First, the linked entities act as seed nodes. Second, a bounded breadth-first expansion collects neighbours along relationship types that are relevant to the intent; for a treatment question the expansion favours TREATS, HAS_CONTRA, INTERACTS_WITH and RECOMMENDS_FOR. Third, passages attached to the collected edges are retrieved through SUPPORTED_BY links, and extra passages are added from dense and lexical search. Fourth, community summaries are added when the question is broad.

Candidates are scored with a weighted combination of graph distance from the seeds, semantic similarity, source authority and recency. Patient-specific filters are applied: contraindications and interactions that match the supplied patient context receive a priority boost so that they cannot be dropped by the context budget.

The relevance score of a candidate evidence item e for query q is computed as:

$$S(e) = \alpha \cdot \text{sim}(q, e) + \beta \cdot \text{proximity}(e) + \gamma \cdot \text{authority}(e) + \delta \cdot \text{recency}(e) + \lambda \cdot \text{patient_match}(e)$$

where α, β, γ, δ and λ are non-negative weights tuned on a validation set, proximity decreases with the number of hops from a seed entity, and patient_match is 1 if the item matches a stated patient attribute and 0 otherwise.

6.10 Module M7: Evidence Assembly and Grounded Prompting

The top-ranked items are arranged into a structured evidence block. Graph facts are rendered as short numbered statements such as "[E3] Drug A TREATS Condition B (source: Guideline X, 2024)", followed by supporting passages and community summaries. The prompt instructs the model to use only the numbered evidence, to cite the evidence number after every statement, to state explicitly when the evidence does not answer the question, to avoid giving personal dosing orders and to mention relevant contraindications present in the evidence. A low sampling temperature is used to reduce variability.

6.11 Module M8: Hallucination Guard and Claim Verification

The hallucination guard is the central safety component. After the draft answer is produced, a claim extractor decomposes it into atomic, self-contained claims. Each claim is then checked in two ways. A structured check tries to match the claim to a graph edge or property, for example (Drug A, TREATS, Condition B), including numeric values and population qualifiers. A textual entailment check compares the claim with the cited passages using a natural language inference model or a verification prompt to an LLM. A claim is labelled supported when it matches an edge or is entailed by a cited passage, contradicted when it conflicts with an edge or passage, and unsupported when neither support nor conflict is found.

Table 6.3: Verification outcomes and system actions

Claim status	Meaning	Action
Supported	Matches a graph fact or is entailed by a cited passage	Keep with citation
Contradicted	Conflicts with the graph or evidence	Remove and regenerate; log as a hallucination event
Unsupported	No evidence found either way	Remove or soften; regenerate with stricter grounding
Safety-critical	Dose, contraindication or	Block release; abstain if not

unsupported	interaction without evidence	resolved

6.12 Confidence Scoring and Abstention

The overall confidence of an answer combines the share of supported claims, the average retrieval score, the authority of the cited sources and the degree of agreement among sources. A simple formulation is:

Confidence = w1 · SupportedRatio + w2 · MeanRetrievalScore + w3 · MeanAuthority + w4 · SourceAgreement

with weights summing to one and calibrated on a validation set so that, for example, answers with confidence 0.8 are correct about 80 percent of the time. The policy engine releases the answer when confidence exceeds an upper threshold and no safety-critical claim failed, adds a caution banner in an intermediate band, and abstains below a lower threshold. An abstention message explains what was missing, so that the clinician knows where to look next.

6.13 Algorithm 1: Knowledge Graph Construction

Input: source set S, ontology O, schema T, thresholds τ_conf, τ_review

Output: validated graph version G

1. for each source s in S do
2. chunks ← Preprocess(s)
3. for each chunk c in chunks do
4. M ← NER(c)
5. R ← LLMExtract(c, M, T)
6. for each triple (h, r, t, sentence, conf) in R do
7. if sentence is not a verbatim span of c then discard
8. h′ ← Link(h, O); t′ ← Link(t, O)
9. if conf ≥ τ_conf then Stage(h′, r, t′, s, sentence, conf)
10. end for
11. end for

12. end for
13. for each staged edge e do
14. if Type(e) is safety-critical and (conf(e) < τ_review or support(e) = 1)
15. then SendToReviewer(e)
16. else Commit(e)
17. end for
18. FlagConflicts(G); ValidateSchema(G)
19. C ← Leiden(G); Summaries ← Summarise(C)
20. EmbedAndIndex(chunks, nodes, Summaries)
21. return Publish(G)

Complexity

Extraction cost is linear in the number of chunks and is dominated by LLM calls, which can be parallelised and cached. Leiden community detection runs in near-linear time in the number of edges for practical graphs. Incremental updates re-process only the changed sources and the communities they touch.

6.14 Algorithm 2: Hybrid Graph Retrieval

Input: question q, patient context P, max hops k, budget B

Output: ranked evidence set E

1. intent ← Classify(q); seeds ← Link(Entities(q), O)
2. frontier ← seeds; F ← seeds
3. for hop = 1 to k do
4. frontier ← Neighbours(frontier, AllowedRelations(intent))
5. F ← F ∪ frontier
6. end for
7. facts ← EdgesBetween(F)
8. passages ← SourcePassages(facts) ∪ DenseSearch(q) ∪ BM25(q)
9. if IsBroad(intent) then summaries ← CommunitySummaries(seeds)
10. for each candidate e in facts ∪ passages ∪ summaries do
11. $S(e) \leftarrow \alpha \cdot sim + \beta \cdot proximity + \gamma \cdot authority + \delta \cdot recency + \lambda \cdot match(e, P)$
12. end for
13. E ← SelectTop(candidates, B) // patient-critical items always retained
14. return E

6.15 Algorithm 3: Verified Answer Generation

Input: question q, evidence E, max attempts m

Output: final response

1. for attempt = 1 to m do
2. draft ← LLM(GroundedPrompt(q, E), temperature = low)
3. claims ← ExtractClaims(draft)
4. for each claim c in claims do
5. status(c) ← Verify(c, E, Graph)

6. end for

7. conf ← Score(claims, E)

8. if no contradicted claim and no safety-critical unsupported claim then

9. if conf ≥ τ_high then return Release(draft, citations, conf)

10. if conf ≥ τ_low then return Release(draft, citations, conf, caution)

11. end if

12. E ← Strengthen(E, failedClaims) // extra retrieval round for failed claims

13. end for

14. return Abstain(reason, partialEvidence)

6.16 Worked Example

Consider the question: "Which antihypertensive options are suitable for a 58-year-old patient with type 2 diabetes and chronic kidney disease who is already taking an ACE inhibitor?" The entity linker identifies hypertension, type 2 diabetes, chronic kidney disease and ACE inhibitor. Graph expansion collects drugs linked to hypertension by TREATS, their contraindications and interactions, and guideline nodes that recommend drug classes for hypertension with diabetic kidney disease. The re-ranker boosts the interaction between an ACE inhibitor and certain other drug classes because the patient already takes an ACE inhibitor. The generator drafts options with citations, and the verifier confirms each option against a guideline edge. A draft statement that a particular combination is "generally safe" is found to have no supporting edge and a conflicting interaction note; the claim is removed and the final answer warns about the interaction. This illustrates how the guard converts a plausible but risky sentence into a safe, cited response.

7. UML DIAGRAMS

The Unified Modeling Language (UML) is used to describe the structure and behaviour of MediGraph AI from several viewpoints. This section presents the use case diagram, class diagram, sequence diagram, activity diagram and deployment diagram. Level 0 DFD (Context Diagram)

7.1 Use Case Diagram

The use case diagram in Figure 7.1 shows the actors and the functions that the system offers to them. Clinicians and medical students interact with the question-answering functions. Knowledge administrators and clinical reviewers maintain the quality of the graph, and the system administrator manages users and monitors operation.

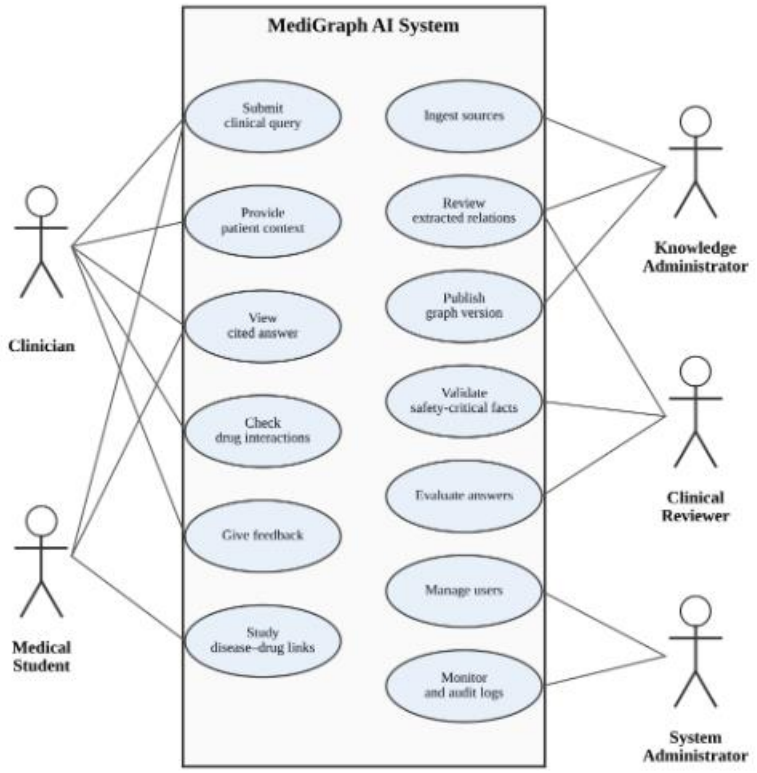


Figure 7.1: Use case diagram of MediGraph AI

Table 7.1: Specification of the main use case "Submit clinical query"

Item	Description
Primary actor	Clinician
Precondition	User is authenticated and the knowledge graph has a published version
Main flow	1. User enters a question and optional patient context. 2. System links entities and retrieves evidence. 3. System generates and verifies the answer. 4. System displays the answer with

	citations, reasoning path and confidence.
Alternative flow	If entities are ambiguous the system asks for clarification. If evidence is insufficient or verification fails repeatedly, the system abstains and explains what was missing.
Postcondition	Query, evidence, claims and decision are recorded in the audit log

7.2Class Diagram

The class diagram in Figure 7.2 shows the main classes, their principal attributes and operations, and their relationships. A QueryRequest is analysed by the QueryAnalyzer, which links entities in the KnowledgeGraph. The HybridRetriever expands the graph and returns EvidenceItem objects, which ground the AnswerGenerator. The generated draft is examined by the ClaimVerifier, which creates Claim objects whose statuses feed the PolicyEngine that decides whether to release, regenerate or abstain

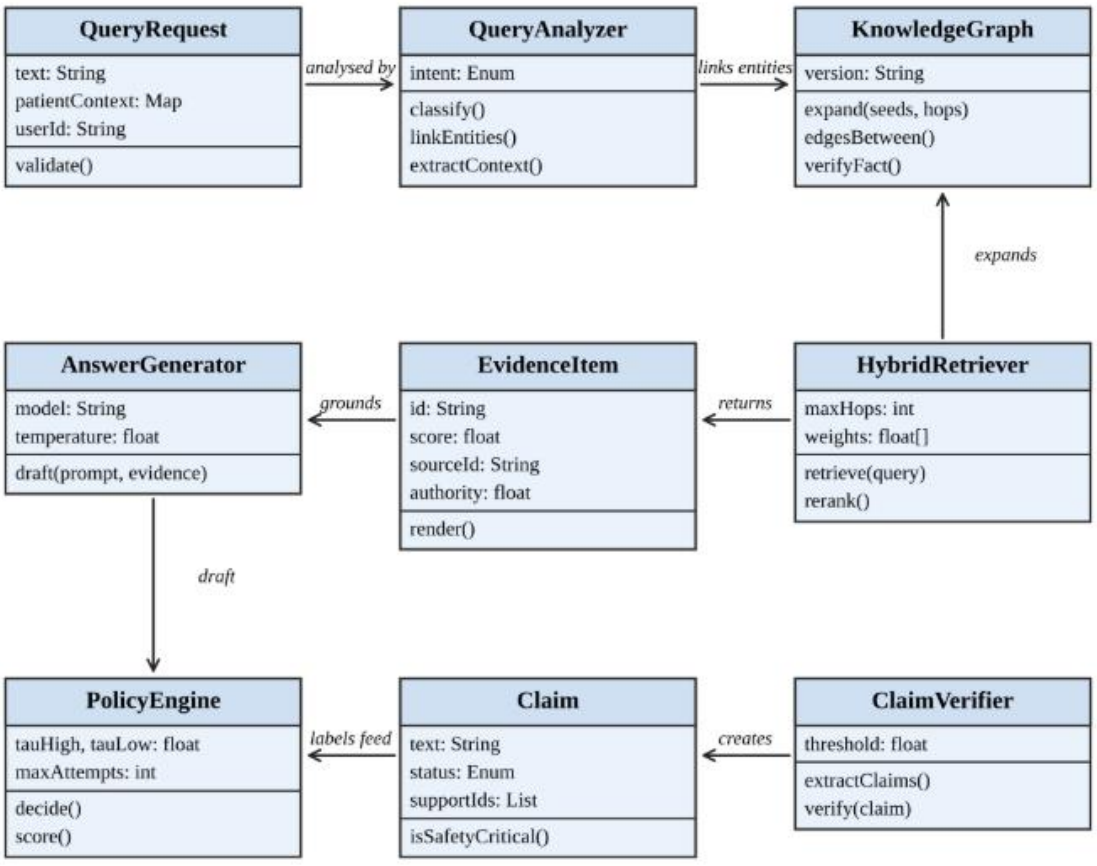


Figure 7.2: Class diagram of MediGraph AI

7.3Sequence Diagram

Figure 7.3 shows the order of messages exchanged during a successful query. The orchestrator coordinates all components: it first obtains evidence through the retriever, then asks the LLM for a draft, and finally submits the draft to the verifier, which consults the knowledge graph before the answer is returned to the user.

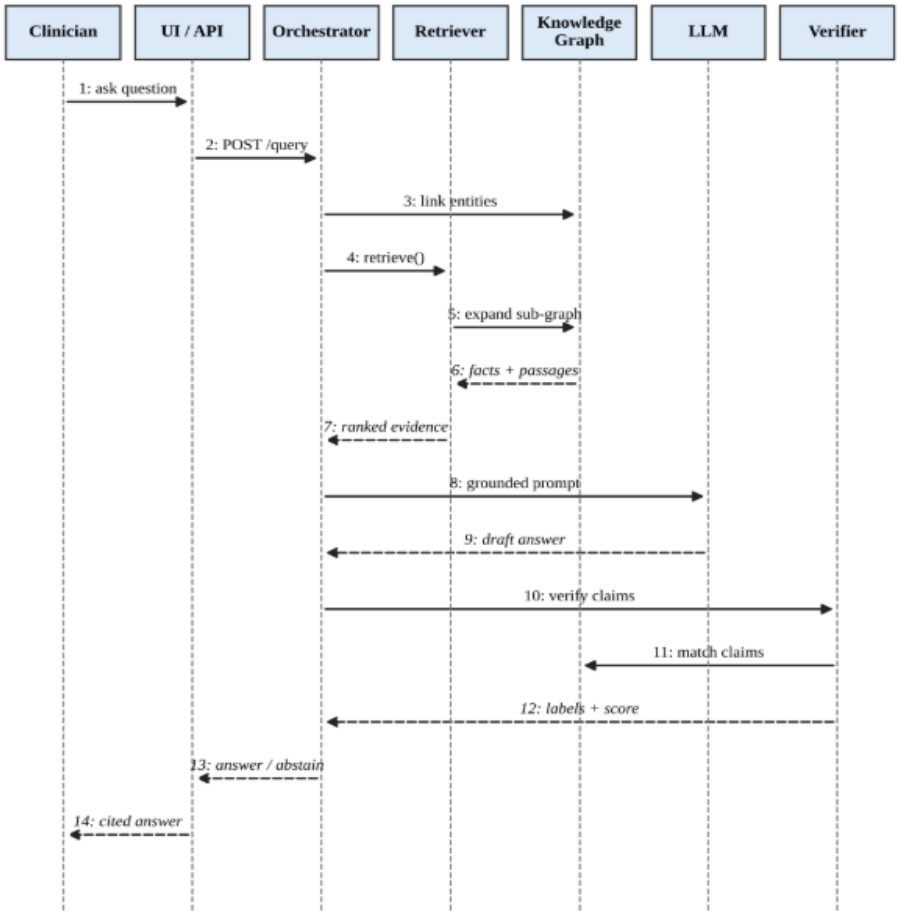


Figure 7.3: Sequence diagram for query processing

7.4 Activity Diagram

The activity diagram in Figure 7.4 describes the control flow, including the decision points that make the system safe. A question with unresolved entities leads to a clarification request. Insufficient evidence leads to abstention. Claims that fail verification trigger a bounded number of regeneration attempts with strengthened evidence, and the confidence score decides whether the answer is released normally or with a caution ba

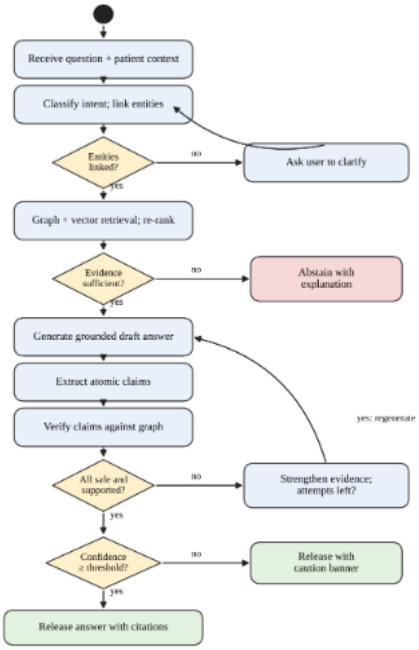


Figure 7.4: Activity diagram of verified answer generation

7.5 Deployment Diagram

Figure 7.5 shows the physical deployment. The browser communicates with an API gateway that performs authentication. The application tier contains the orchestrator, retriever and verifier as containerised services. The data tier holds the graph database, the vector and lexical index and the relational database. Model services provide LLM, embedding and entailment inference, and the offline ingestion workers update the data tier from raw sources held in object storage.

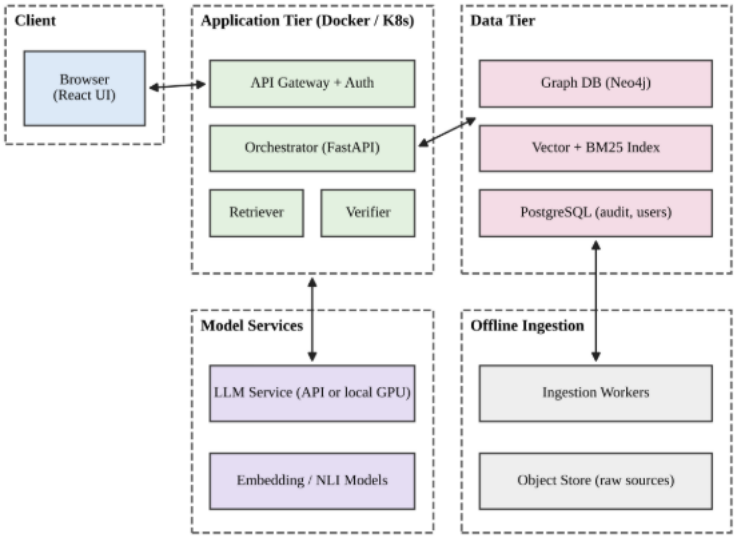


Figure 7.5: Deployment diagram of MediGraph AI

8. REFERENCES

[1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. https://arxiv.org/abs/2005.11401

[2] D. Edge et al., "From Local to Global: A Graph RAG Approach to Query-Focused Summarization," 2024. https://arxiv.org/abs/2404.16130

[3] J. Wu et al., "Medical Graph RAG: Towards Safe Medical Large Language Model via Graph Retrieval-Augmented Generation," 2024. https://arxiv.org/abs/2408.04187

[4] J. Wu et al., "Medical Graph RAG: Evidence-Based Medical Large Language Model via Graph Retrieval-Augmented Generation," *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. https://aclanthology.org/2025.acl-long.1381/

[5] K. Singhal et al., "Large Language Models Encode Clinical Knowledge," *Nature*, vol. 620, pp. 172–180, 2023. https://doi.org/10.1038/s41586-023-06291-2

[6] H. Nori et al., "Capabilities of GPT-4 on Medical Challenge Problems," 2023. https://arxiv.org/abs/2303.13375

[7] L. Huang et al., "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," 2023. https://arxiv.org/abs/2311.05232

[8] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems*, 2020. httpsA://arxiv.org/abs/2005.11401

[9] C. Shu et al., "ClinicalRAG: Enhancing Clinical Decision Support through Heterogeneous Knowledge Retrieval," 2024.https://aclanthology.org/2024.knowllm-1.6/

[10] World Health Organization, "Ethics and Governance of Artificial Intelligence for Health: WHO Guidance," World Health Organization, 2021. https://www.who.int/publications/i/item/9789240029200

[11] National Library of Medicine, "Unified Medical Language System (UMLS)," National Institutes of Health. https://www.nlm.nih.gov/research/umls/

[12] International Health Terminology Standards Development Organisation, "SNOMED CT: Systematized Nomenclature of Medicine Clinical Terms." https://www.snomed.org/

[13] National Library of Medicine, "Medical Subject Headings (MeSH)," National Institutes of Health. https://www.nlm.nih.gov/mesh/

[14] National Library of Medicine, "PubMed: A Database of Biomedical Literature," National Institutes of Health. https://pubmed.ncbi.nlm.nih.gov/

[15] World Wide Web Consortium (W3C), "OWL 2 Web Ontology Language Document Overview," W3C Recommendation, 2012. https://www.w3.org/TR/owl2-overview/

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Review

A Survey on MLLMs in Education: Application and Future Directions

Weicheng Xing, Tianqing Zhu, Jenny Wang and Bo Liu

Special Issue

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Review

A Survey on MLLMs in Education: Application and Future Directions

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Abstract: This survey paper examines the applications, methodologies, and future prospects of multimodal large language models (MLLMs) within the educational landscape. MLLMs, which integrate multiple data modalities such as text, images, and audio, offer innovative solutions that enhance learning experiences across various educational domains, including language acquisition, STEM education, interactive content creation, and medical training. The paper highlights how MLLMs contribute to improved engagement, personalized learning paths, and enhanced comprehension by leveraging their ability to process and generate contextually relevant content. The key findings underscore the transformative potential of MLLMs in modern education, suggesting significant improvements in both learner outcomes and pedagogical strategies. The paper also explores emerging trends and technological advancements that could shape the future of education, advocating for continued research and collaboration among stakeholders to fully harness the capabilities of MLLMs. As the integration of MLLMs into educational settings progresses, addressing ethical considerations and ensuring equitable access remain critical to maximizing their benefits.

Keywords: multimodal large language models (MLLMs); AI in education; educational technology (EdTech); multimodal integration in learning; computer vision in education

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1. Introduction

The integration of technology in education has undergone significant evolution over the past few decades, transforming learning landscapes and pedagogical practices. From early computer-assisted learning programs to sophisticated AI-driven educational applications, technology's impact on education has been profound and multifaceted [1,2]. These innovations have provided educators and learners with unprecedented access to information, personalized learning experiences, and tools that enhance engagement and comprehension.

Recently, artificial intelligence (AI) has emerged as a pivotal force in education, revolutionizing how content is delivered and consumed. AI-driven educational applications, such as adaptive learning systems and intelligent tutoring, have demonstrated the potential to tailor educational experiences to individual learner needs, promoting greater accessibility and inclusivity [3]. The historical context of these technologies highlights their significance in addressing diverse educational challenges, from bridging knowledge gaps to fostering critical thinking skills.

Among the most transformative developments in AI are large language models (LLMs), such as OpenAI's GPT-4o and GPT-o1, which have exhibited remarkable capabilities in understanding and generating human-like text [4]. Trained on vast amounts of data, these models can perform a wide array of language tasks, including translation, summarization, question-answering, and creative writing. The introduction of LLMs into

educational settings has opened new avenues for enhancing teaching and learning processes. They offer personalized feedback, generate educational content, support language learning, and even assist in administrative tasks [1].

Advancing beyond LLMs, multimodal large language models (MLLMs) extend these capabilities by integrating multiple data modalities such as text, images, audio, and video [5]. This integration enables MLLMs to process and generate content across different formats, making them particularly valuable for educational applications that require a holistic understanding of multimodal information. For instance, MLLMs can assist in creating interactive learning materials, facilitating visual explanations of complex concepts, and supporting accessibility for learners with diverse needs [6].

The convergence of LLMs and multimodal processing represents a significant shift in educational technology, promising to further personalize and enrich learning experiences. As these models become increasingly sophisticated and accessible, understanding their potential applications and implications in education becomes crucial.

This survey aims to explore the current state of MLLMs in education, examining their applications, challenges, and future directions. The main contributions of this paper are threefold: (1) an exploration of the current applications of MLLMs in education, highlighting their potential to transform traditional learning methods; (2) an analysis of the challenges and ethical considerations associated with implementing MLLMs in educational environments, such as data privacy and security; and (3) a discussion of emerging trends and future directions for MLLMs in education [7], with recommendations for further research and collaboration.

The overall structure of the survey can be viewed in Figure 1. The Introduction sets the stage by placing multimodal large language models (MLLMs) into an educational context, examining their growing importance in teaching and learning. Preliminaries of MLLMs then offers a foundational exploration of their core components and underlying technologies, clarifying how they integrate multiple modalities. Applications of MLLMs in Education illustrates their current roles in enhancing adaptive learning, providing virtual tutoring, automating content creation, and streamlining educational management systems. Discussion and Future Directions addresses ethical issues, technological hurdles, and the need for improved personalization, engagement, and accessibility, suggesting pathways to strengthen MLLMs' contributions to the educational landscape. Finally, Conclusions synthesizes the key insights and findings, highlighting the paper's main contributions and implications for future practice and research.

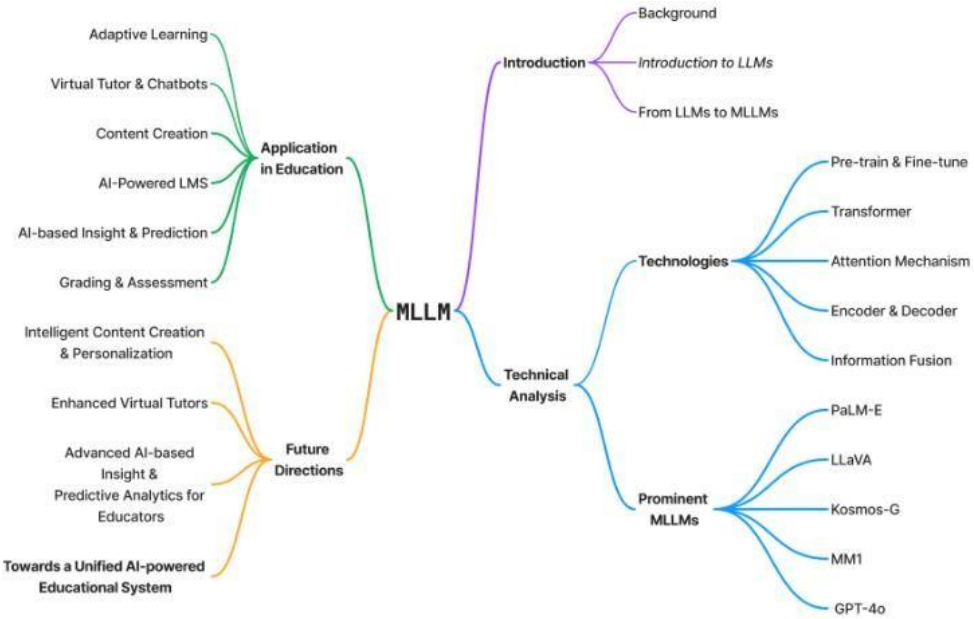


Figure 1. The overall structure of the survey.

2. Preliminaries of MLLMs

2.1. Overview of Large Language Models (LLMs)

Large language models (LLMs) have revolutionized the field of artificial intelligence, particularly in natural language processing (NLP) tasks. These models are designed to understand, generate, and manipulate human language with a high degree of fluency and coherence. Notable examples include OpenAI's GPT-4o and Anthropic's Claude3.5, which are trained on vast amounts of textual data and can perform a wide array of language-related tasks such as translation, summarization, question answering, and content creation [8–10]. LLMs leverage the transformer architecture [11], utilizing self-attention mechanisms to capture complex dependencies in language, enabling them to generate human-like text that is contextually relevant.

The transformer architecture has been pivotal in the success of LLMs. It allows models to process input data in parallel rather than sequentially, significantly improving computational efficiency and performance on large datasets. Self-attention mechanisms within transformers enable the models to weigh the importance of different parts of the input data, capturing long-range dependencies and nuances in language [6]. This capability has led to breakthroughs in tasks that were previously challenging for AI, such as coherent essay writing and intricate dialogue generation.

Despite their impressive capabilities, traditional LLMs are limited to processing textual data [12]. This constraint restricts their applicability in contexts where information is conveyed through multiple modalities, such as images, audio, and video. Human communication and learning are inherently multimodal, often involving the integration of visual cues, auditory signals, and textual information. The inability of LLMs to process non-textual data limits their effectiveness in applications that require a holistic understanding of diverse information sources.

2.2. Evolution to Multimodal Large Language Models (MLLMs)

To address the limitations of traditional LLMs, researchers have developed multimodal large language models (MLLMs), which extend the abilities of LLMs by integrating and processing diverse data modalities alongside text [6,12,13]. MLLMs represent a significant advancement in AI, enabling the fusion of textual and non-textual data to create models that can comprehend and generate contextually relevant outputs across different modalities. This integration allows MLLMs to perform complex tasks that mirror human-like understanding and communication, such as interpreting images with accompanying descriptions or generating detailed visual content from textual prompts.

The development of MLLMs has led to several notable models that demonstrate the seamless integration of multiple modalities. One such model is OpenAI's CLIP (Contrastive Language–Image Pre-training), which learns visual concepts from natural language supervision [14]. CLIP aligns images and textual descriptions in a shared embedding space, allowing it to perform tasks such as image classification, object recognition, and image-text retrieval without explicit task-specific training data. By leveraging a contrastive learning approach, CLIP can understand the relationship between images and text, making it highly versatile in handling multimodal data.

Similarly, DALL·E and DALL·E 2 are models capable of generating high-quality images from textual prompts by utilizing diffusion models conditioned on text embeddings [15,16]. These models showcase the ability to merge text and visual understanding seamlessly, creating novel images that correspond to detailed textual descriptions. This capability has profound implications for creative industries, design, and educational content creation, where visualizations generated from textual concepts can enhance comprehension and engagement.

Another significant development is DeepMind's Flamingo, a visual language model that can perform few-shot learning across multiple modalities [13]. Flamingo integrates vision and language modalities in a single model, enabling it to handle tasks like visual question answering, image captioning, and dialogue with minimal task-specific training. By

leveraging large-scale pre-training and attention mechanisms [17], Flamingo can adapt to new tasks with limited data, highlighting the potential for efficient and scalable deployment of MLLMs in various applications.

In educational contexts, MLLMs support diverse learning needs by facilitating adaptive and inclusive educational experiences. For example, they can generate interactive learning materials, produce visual explanations for complex topics, and provide accessibility features, such as audio descriptions for visual content. The transition from text-based LLMs to MLLMs thus marks a significant shift in educational technology, offering educators and students new opportunities to interact with AI in more dynamic, immersive, and effective ways [18].

As MLLMs continue to evolve, they will further redefine the educational landscape, enabling a deeper level of personalization, engagement, and accessibility in learning. This integration of multiple modalities aligns well with the diverse learning styles and needs of today's students, enhancing both teaching and learning experiences.

2.3. Key Technologiesand Architectures

The development of multimodal large language models (MLLMs) relies on several advanced technologies and architectures, each playing a crucial role in enabling these models to process and integrate data from multiple modalities effectively. Figure 2 shows the general model architecture of MLLMs. This subsection highlights the core components, including the transformer architecture, fusion techniques, and specialized training methods that empower MLLMs to generate coherent and contextually relevant multimodal outputs.

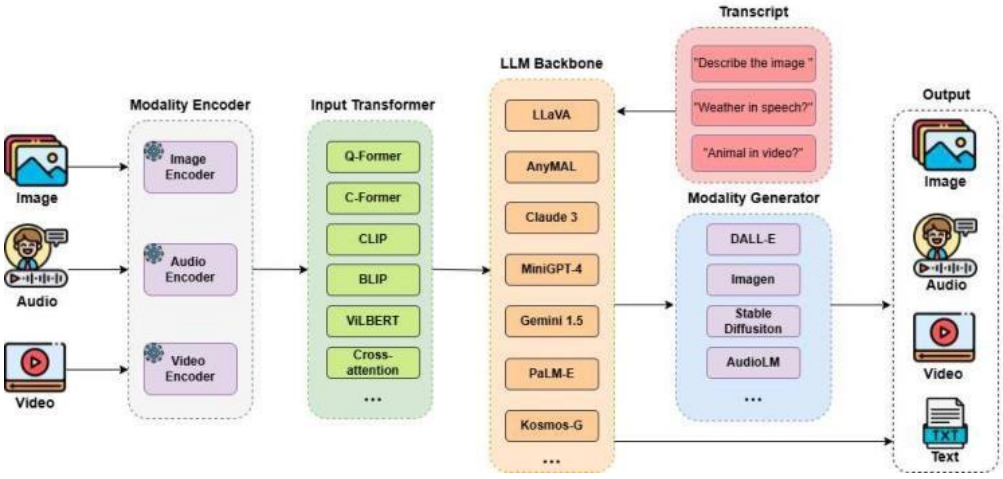


Figure 2. The general model architecture of MLLMs.

2.3.1. Transformer Architectures and Attention Mechanisms

At the core of MLLMs are neural networks based on the transformer architecture [6], which are well suited for handling sequential data and learning complex patterns across different modalities. Transformers use self-attention mechanisms to weigh the significance of different parts of the input data, allowing the model to focus on the most relevant information [19]. Extensions of the transformer architecture have been developed to handle multimodal inputs, such as Vision Transformer (ViT) for images [20] and Audio Transformer for audio data [21].

Attention mechanisms are crucial for MLLMs, enabling them to prioritize certain aspects of input data over others. In multimodal settings, cross-modal attention allows the model to align and fuse information from different modalities [22]. For example, in visual question-answering tasks, the model needs to attend to relevant regions in an image based on a textual question [23].

2.3.2. Multimodal Fusion Techniques

Information fusion is a critical component in MLLMs, enabling the effective combination of data from different modalities. Fusion methods are typically categorized into the following categories:

- **Early fusion (single-stream)**: Integrates modalities at the input level, allowing the model to learn cross-modal representations from the beginning. This method processes all modalities simultaneously, enabling the model to capture interactions between different types of data early in the processing pipeline [6].
- **Late fusion (dual-stream)**: Processes each modality separately before combining their representations at a higher level. This approach allows for specialized processing of each modality, which can be beneficial when modalities have very different characteristics or when pre-trained unimodal models are used [6].

The choice between early and late fusion depends on the specific application and the nature of the data involved. Effective fusion strategies are essential for MLLMs to leverage the complementary information present in different modalities, leading to more robust and accurate models.

2.3.3. Pre-Training and Fine-Tuning

MLLMs often undergo extensive pre-training on large multimodal datasets to acquire a broad understanding of various data types [5,24]. This is followed by fine-tuning on specific tasks or domains to optimize their performance for particular applications [25–27].

2.3.4. Encoder and Decoder Architectures

The architecture of MLLMs often builds upon the transformer framework, extending it to handle multiple modalities. Two primary architectural approaches are used:

- **Encoder-only models**: Models like CLIP focus on creating embeddings for different modalities that can be compared or combined [14]. The encoder processes input data to generate a fixed-size representation, capturing the essential features of the input regardless of its modality. This approach is effective for tasks that require matching or retrieving information across modalities.
- **Encoder–decoder models**: Models used in tasks like image captioning process input data through an encoder and generate outputs via a decoder, allowing for generative tasks [5]. The encoder transforms the input data into a latent representation, which the decoder then uses to generate a sequence of outputs in another modality. This architecture is well suited for tasks that involve translation between modalities, such as generating descriptive text from images.

2.4. Examples of Prominent MLLMs

As shown in Figure 3, several notable multimodal large language models (MLLMs) have emerged in recent years, showcasing the transformative potential of this technology in various fields.

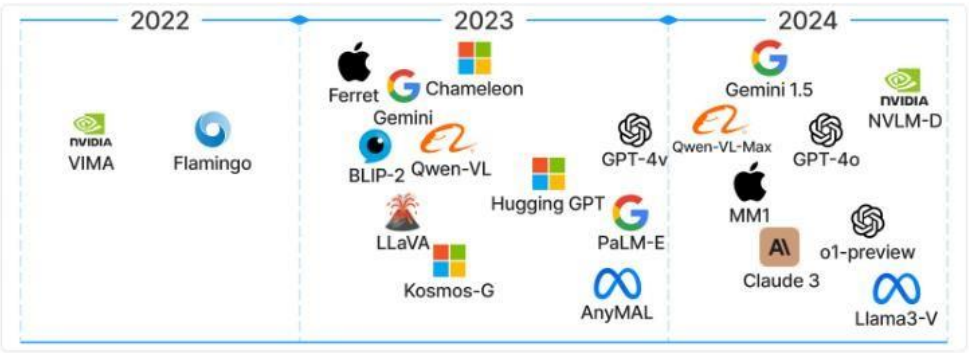


Figure 3. Timeline of major representative MLLMs, starting from 2022.

We selected some of the state-of-the-art MLLMs to analyze their technical details as well as their advantages and disadvantages in application. It is worth noting that we found the open-source MLLM technical information to be more detailed, allowing us to perform a more nuanced analysis. Table 1 provides an overview of some of the most impactful MLLMs and their key features.

Table 1. Overview of selected prominent MLLMs and their educational applications.

Model	Key Capabilities	Educational Applications	Open Source
PaLM-E	Integrates vision and language for robotics and embodied AI.	Enhances interactive, physical learning environments, especially in STEM and robotics education.	No
LLaVA	Combines vision and language for general-purpose understanding.	Visual question answering, image captioning, supporting visually enriched content in learning platforms.	Yes
Kosmos-G	Processes text and images for multimodal comprehension.	Facilitates interactive content, supports collaborative and visual learning tools.	No
GPT-4o	Extends LLM capabilities to visual data, enabling conversational responses to images.	Supports interactive learning with text and visual input, such as image-based Q&A and description generation.	No
MM1	Achieves state-of-the-art performance in multimodal tasks by combining high-resolution visual processing and language models.	Supports tasks like in-context learning, multi-image reasoning, and few-shot learning, useful for assessments and exploratory learning.	No
Llama 3-V	Combines vision, language, coding, reasoning, and tool usage with multilingual support.	Enables adaptive and multilingual learning, coding education, and collaborative projects.	Yes
NVLM 1.0	Frontier-class multimodal model with exceptional vision-language reasoning and text-only improvements.	Supports OCR tasks, multimodal math reasoning, and document analysis in educational environments.	Yes
BLIP	Pre-trained for text-image retrieval and multimodal content generation.	Facilitates creative content development and storytelling in visual education.	Yes

2.4.1. Open-Source MLLMs

NVLM 1.0: Developed by NVIDIA, NVLM 1.0 is a frontier-class family of multimodal large language models designed to excel in vision-language tasks while maintaining or even improving text-only performance [28]. NVLM introduces three architectural variants: the decoder-only NVLM-D, the cross-attention-based NVLM-X, and the hybrid NVLM-H, each catering to different multimodal processing needs. With model sizes up to 72 billion parameters, NVLM combines state-of-the-art performance with high flexibility for handling diverse multimodal inputs.

NVLM's training process incorporates a meticulously curated dataset blend, including high-quality multimodal and text-only supervised fine-tuning (SFT) datasets. This approach preserves text-only capabilities while enhancing vision-language performance. NVLM-D processes multimodal input directly within the LLM, achieving unified reasoning capabilities. NVLM-X employs gated cross-attention for efficient high-resolution image processing, while NVLM-H combines elements of both architectures, providing superior reasoning and computational efficiency.

One of NVLM 1.0's key strengths is its performance on diverse benchmarks, where it rivals or surpasses proprietary models like GPT-4o and Claude 3.5. For instance, NVLM-D demonstrates exceptional OCR capabilities, achieving the highest scores on benchmarks such as OCRBench and VQAv2. NVLM-H leads in multimodal reasoning tasks like MathVista and multidisciplinary reasoning, proving its versatility for complex educational applications.

Despite these advancements, NVLM models also present challenges. The large-scale models require substantial computational resources for both training and deployment,

which may limit accessibility for smaller institutions. Additionally, while NVIDIA plans to open-source the training code and model weights, the complexity of implementation might pose challenges for non-specialist users without significant expertise or infrastructure.

NVLM 1.0's public release is a significant contribution to the open-source ecosystem, offering powerful tools for research and development in multimodal AI. Its focus on maintaining text-only performance during multimodal training sets a new standard, enabling its integration into educational technologies without compromising existing capabilities. By providing production-grade multimodality and robust benchmarks, NVLM supports the creation of advanced applications across various fields, including education, healthcare, and content generation.

Potential applications: NVLM 1.0 can be utilized to build OCR-based educational tools, multimodal math reasoning assistants, and adaptive content generation systems. For example, its superior OCR capabilities can support the digitization of historical documents for research purposes, while its multimodal reasoning abilities can enhance STEM education with visual problem-solving tasks.

Llama 3-V: Llama 3-V is the latest addition to Meta's family of open-source foundation models, featuring a herd of models designed to natively support multilinguality, coding, reasoning, and tool usage. The largest variant of Llama 3-V is a dense transformer model with 405 billion parameters and a context window of up to 128 K tokens [29]. The models are trained on an unprecedented corpus of 15 trillion multilingual tokens, enabling them to achieve state-of-the-art performance across a wide range of language understanding tasks. The Llama 3-V models, available under an updated open-source license, include pre-trained and post-trained versions, with the flagship Llama 3-V.1 model delivering competitive quality compared to GPT-4.

One of the major advantages of Llama 3-V lies in its versatility and performance. By supporting multilinguality and tool usage, it offers immense flexibility for global educational applications, including language instruction, cross-cultural studies, and adaptive learning environments. Its extended context window makes it particularly well suited for tasks requiring in-depth reasoning, such as literature analysis, historical document interpretation, and coding exercises. Additionally, Meta's emphasis on safety, demonstrated through the Llama Guard 3 model for input and output safety, ensures that the models can be deployed responsibly in educational contexts.

Despite its strengths, Llama 3-V presents challenges. While being open-source, its large-scale models like the 405B parameter variant require substantial computational resources, making them difficult to deploy for institutions with limited infrastructure. The extensive training and fine-tuning processes, although yielding exceptional results, also necessitate expertise and investment in hardware and personnel. Moreover, while Meta has improved pre-training data quality and post-training alignment processes, the scale of Llama 3-V may raise concerns about data privacy and ethical implications in specific use cases.

Llama 3-V's compositional approach to multimodality—integrating image, video, and speech recognition capabilities—shows promising results that rival state-of-the-art models in these domains. Although these multimodal extensions are still under development, they highlight the potential of Llama 3-V to transform education by enabling innovative applications such as visual storytelling, multilingual video-based lessons, and speech-assisted learning tools. The commitment to public release ensures transparency and fosters a collaborative research ecosystem to enhance the model further.

Overall, Llama 3-V represents a significant milestone in the evolution of open-source MLLMs, demonstrating the potential of large-scale, multimodal models to advance global education and research. Its balance between performance, openness, and safety positions it as a valuable asset for institutions aiming to integrate cutting-edge AI technologies into their educational strategies.

Potential applications: Llama 3-V can enable visual storytelling, multilingual video-based lessons, and adaptive learning platforms. Its large context window is particularly

useful for creating tools that assist in complex problem solving, such as code debugging environments or interactive literature exploration.

LLaVA (Large Language and Vision Assistant): LLaVA is an open-source model designed to combine visual encoders with large language models, enabling general-purpose visual and language understanding [30]. It supports tasks such as visual question answering, image captioning, and the development of interactive, multimodal educational content. One major advantage of LLaVA is its accessibility and adaptability. Being open-source, it allows educators and developers to customize the model to address specific learning needs, such as creating tailored visual aids or interactive lessons. Furthermore, it fosters innovation by providing a platform for experimentation and collaboration within the educational community.

LLaVA also poses significant challenges. The implementation of the model requires considerable technical expertise and computational resources, which may not be readily available to many educational institutions. Furthermore, the absence of dedicated support services means that troubleshooting and model optimization must often be handled internally, potentially delaying adoption and reducing its usability in time-sensitive scenarios.

Potential applications: LLaVA can support the creation of interactive learning environments, such as visual science tutorials or personalized art history lessons. Its adaptability makes it ideal for generating tailored visual aids for diverse subjects.

2.4.2. Proprietary MLLMs

GPT-4o (Multimodal Version): GPT-4o, developed by OpenAI, integrates visual and textual data for advanced multimodal reasoning. Its applications in education are vast, ranging from interactive learning environments to visual question answering and conversational image interpretation. One of its primary advantages is its exceptional performance and reliability [31]. The model's robust capabilities ensure accurate and consistent results across a wide range of tasks, making it a reliable tool for educators. Moreover, the professional support and regular updates provided by OpenAI enhance its usability and stability in educational settings, ensuring that institutions can deploy the model with confidence.

GPT-4o's proprietary nature introduces challenges. The high licensing fees can be prohibitive for many educational institutions, particularly those operating on limited budgets. Additionally, its closed design restricts customization, preventing institutions from tailoring the model to specific educational contexts or integrating it with existing systems beyond the predefined parameters.

Potential applications: GPT-4o can be used to create advanced virtual tutors capable of visual and textual reasoning. For example, it could support biology lessons with real-time microscopic image analysis or assist in art classes with detailed image composition feedback.

Kosmos-G: Developed by Microsoft, Kosmos-G offers enhanced multimodal capabilities, including advanced visual and text comprehension. It is well suited for collaborative and visual learning tools, integrating seamlessly with Microsoft's ecosystem, such as Teams and Office 365, to provide a unified platform for education [32]. This integration simplifies its adoption and enhances user experience, particularly for institutions already utilizing Microsoft products.

Nevertheless, Kosmos-G's advantages are accompanied by significant limitations. The model's cost can be a major barrier for adoption, as it requires both licensing fees and infrastructure investment. Moreover, its reliance on Microsoft's ecosystem may limit its flexibility, making it less suitable for institutions that use diverse or non-Microsoft technologies.

Potential applications: Kosmos-G can enhance educational platforms like Teams by integrating multimodal features, such as document summarization and diagram understanding, making collaborative projects more efficient.

MM1: MM1 is a family of multimodal large language models (MLLMs) developed by Apple, designed to achieve state-of-the-art performance in multimodal tasks by carefully analyzing architectural components and data choices [33]. MM1 includes both dense models with up to 30 billion parameters and mixture-of-experts (MoE) models with up to 64 billion parameters. Through comprehensive ablation studies, MM1 identifies crucial design principles for integrating visual and textual data, such as the importance of image resolution, visual encoder capacity, and the mix of pre-training data. The models exhibit enhanced capabilities like in-context learning, multi-image reasoning, and few-shot chain-of-thought prompting. MM1 demonstrates competitive performance across a range of established multimodal benchmarks, highlighting the effectiveness of large-scale multimodal pre-training and offering valuable insights for future MLLM development.

MM1's proprietary status limits its accessibility. The high costs associated with its deployment, both in licensing and required infrastructure, restrict its use to well-funded institutions. Additionally, its closed system design precludes customization, which may limit its adaptability for diverse educational needs.

Potential applications: MM1 can support advanced assessments, such as AI-driven exam proctoring or complex data interpretation tasks in STEM fields, making it a valuable tool for high-level education.

3. Applications of MLLMs in Education

The advent of multimodal large language models (MLLMs) has ushered in a new era in educational technology, offering innovative solutions that enhance both teaching and learning experiences. Using the ability to process and generate content across multiple modalities, such as text, images, audio, and video, MLLMs are transforming the educational landscape [34]. This section explores the diverse applications of MLLMs in education, highlighting how these advanced AI models are being integrated into various educational tools and platforms to address longstanding challenges and meet the evolving needs of learners and educators.

Table2 summarizes several key areas where MLLMs are making a significant impact:

- **Adaptive learning platforms**: Examining how MLLMs enable personalized learning experiences by dynamically adjusting instructional content to meet individual learners' needs, preferences, and performance levels.
- **Virtual tutors and chatbots**: Exploring the role of MLLMs in developing intelligent virtual assistants that provide personalized support, guidance, and interactive learning opportunities through natural language conversations.
- **Intelligent content creation**: Investigating how MLLMs automate and enhance the development of educational materials, including textbooks, lesson plans, assessments, and multimedia resources, thereby increasing efficiency and accessibility.
- **AI-powered learning management systems (LMSs)**: Analyzing the integration of MLLMs into LMS platforms to enhance content delivery, personalize learning paths, and facilitate more engaging and interactive educational experiences.
- **AI-based insight and predictive analytics for educators**: Discussing how MLLMs provide educators with actionable insights by analyzing multimodal educational data, enabling early identification of at-risk students and informed decision making.
- **Grading and assessment tools**: Assessing the application of MLLMs in automating grading processes, providing objective evaluations, and delivering detailed, personalized feedback across various types of student work.

Through these explorations, the section aims to demonstrate the transformative potential of MLLMs in education. By presenting case studies, technological insights, and practical applications, we seek to highlight how MLLMs not only enhance learning outcomes and engagement but also support educators in delivering high-quality, personalized instruction. The objective is to provide a comprehensive understanding of how MLLMs are shaping the future of education and to inspire further research and implementation in this promising field.